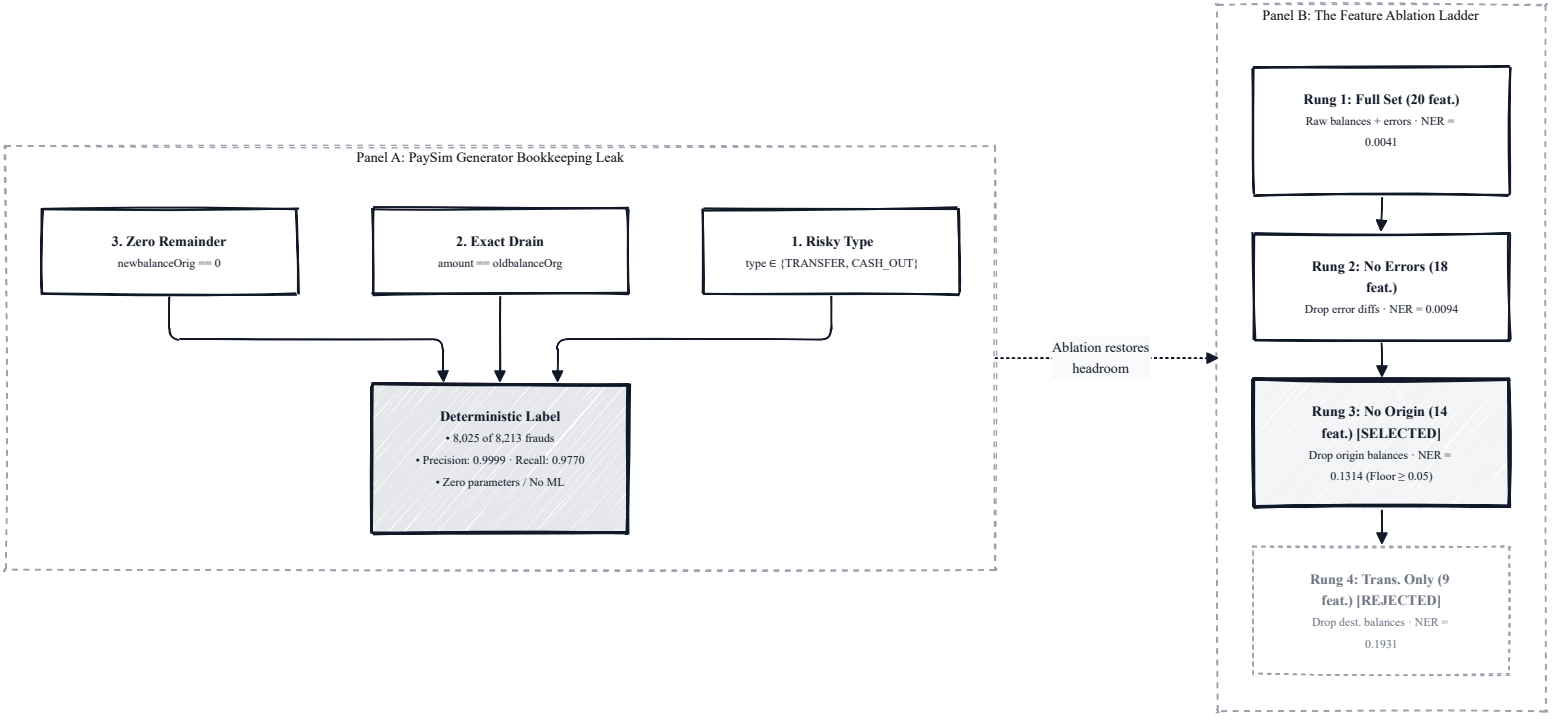

Panel A: PaySim Generator Bookkeeping Leak
3. Zero Remainder
newbalanceOrig == 0
2. Exact Drain
amount == oldbalanceOrg
1. Risky Type
type ∈ {TRANSFER, CASH_OUT}
Deterministic Label
• 8,025 of 8,213 frauds
• Precision: 0.9999 · Recall: 0.9770
• Zero parameters / No ML
Ablation restores headroom
Panel B: The Feature Ablation Ladder
Rung 1: Full Set (20 feat.)
Raw balances + errors · NER = 0.0041
Rung 2: No Errors (18 feat.)
Drop error diffs · NER = 0.0094
Rung 3: No Origin (14 feat.) [SELECTED]
Drop origin balances · NER = 0.1314 (Floor ≥ 0.05)
Rung 4: Trans. Only (9 feat.) [REJECTED]
Drop dest. balances · NER = 0.1931